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Review article

Modeling natural neural networks of decision making with artificial neural networks

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ABSTRACT

One main focus in neuroscience is to understand the relationship between decision making and various brain regions. Researchers use machine learning approaches to model the neural circuits of cerebral cortices, cerebellum, and basal ganglia. This review focuses on artificial neural networks (ANNs), particularly recurrent neural networks (RNNs), to model cortical functions for decision making. We first introduce the basic architecture of RNNs and explain how researchers compare the activity and circuits between artificial and biological networks. We also summarize how RNNs model the prefrontal and posterior parietal cortical in tasks involving short-term memory, perceptual decision making, and value-based decision making. We then show our recent challenges to develop a real-cyber hybrid network, that integrates neuronal activity in mice with RNN-based artificial units to better generate continuous-time body movements, compared to conventional RNNs that only use artificial units. The hybrid network tries to develop RNNs which have similar activity to the brain by using real neurons, rather than developing artificial RNNs and comparing their functions with biological brain. We propose that such integrative approaches in neuroscience and AI will further our understanding of both natural and artificial intelligence in the field of neuro-AI.

1. Introduction

One major focus in systems neuroscience is understanding the brain's learning architecture across different regions (Dayan and Daw, 2008; Doya, 2008, 1999; Perich and Rajan, 2020). Many researchers have used artificial intelligence (AI) to model the neural algorithm of decision-making by comparing behavior, neural circuits, and activity between the artificial and biological systems. In this review, the term "AI" refers to general machine learning approaches, including supervised, unsupervised, and reinforcement learning (RL) (Schmidhuber, 2015). This review mainly focuses on the specific methods in AI, artificial neural networks (ANNs), to summarize how researchers employ them to model the neuronal circuits for decision-making.

Following recent advances in ANNs (Schrittwieser et al., 2020; Silver et al., 2016), there has been a growing interest in employing them to model various brain functions. This review is inspired by the field of neuro-inspired AI (neuro-AI), which has recently gained attention in both AI and neuroscience (Hassabis et al., 2017; Luppi et al., 2024; Richards et al., 2022,2019; Sadeh and Clopath, 2025; Saxe et al., 2021; Zador et al., 2023). The neuro-AI uses the knowledge of neuroscience to develop AI architectures (Ohmae and Ohmae, 2024), similar to how RL and deep learning are developed from biological observations (Fukushima, 1980; Krizhevsky et al., 2017; Sutton and Barto, 2018).

The early history of ANN is well summarized by the scientific background of the Nobel Prize in Physics in year 2024 ("Nobel Prize in Physics, 2024"), which highlights key breakthroughs from the 1980s, including the feedforward and recurrent neural networks. This progress included the "Neocognitoron", inspired by neural activity of the primary visual cortex in cat (Fukushima, 1980), and important theoretical developments (Amari, 1993, 1972). Early efforts to connect ANNs with neural circuits mainly centered on the visual cortex, including the convolutional neural networks (CNNs) (Lecun et al., 1998) and autoencoders (Le et al., 2011). Using deep, multilayered CNNs, researchers modeled the hierarchical structure of visual cortex for object

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identification (Cichy et al., 2016; Yamins et al., 2014; Yamins and DiCarlo, 2016).

In parallel with neural-circuit modeling of the visual cortices using ANNs, the circuit modeling of action selection (i.e., decision making) has also advanced (Fascianelli et al., 2024; Lo and Wang, 2006; Mante et al., 2013; Wang, 2002). This review highlights recurrent neural networks (RNNs) that model the representations of cerebral cortex involved in sensory processing (Shymkiv et al., 2025), short-term memory (Brunel, 1996; Brunel and Wang, 2001; Ji-An et al., 2025), perceptual decision making (Carnevale et al., 2015; Wang, 2002) and value-based decision making (Jensen et al., 2024; Kay et al., 2024; Wang et al., 2018). RNNs have also been employed to perform multiple tasks within a single network (Driscoll et al., 2017; Tafazoli et al., 2024; Yang et al., 2019). We begin by describing the fundamental features of RNNs and introducing an approach for comparing circuits between biological brain and ANNs. We then summarize how RNNs implement cortical circuits for decision making, focusing on the prefrontal cortex (PFC) (Goudar et al., 2023; Jahn et al., 2024), and posterior parietal cortex (PPC) (Cross et al., 2021), which are essential for flexible decision makings (Durstewitz et al., 2023). Lastly, we introduce our recent challenge to develop a real-cyber hybrid network integrating the neurons in animals and artificial units in AI, aiming to develop a brain-like AI compared to conventional AIs with only artificial units.

2. Basics of recurrent neural networks (RNNs)

There are broadly two classes in mathematical models of neural networks, based on the level of biophysical details one wishes to capture. One is a spiking neural network, which explicitly models membrane-potential dynamics; the other is a more coarse-grained, firing-rate-based model. Here, we focus primarily on the rate-based model because it is useful for understanding basic properties of dynamics and learning of neural networks and can achieve, in machine learning, excellent performance across a variety of tasks (Fig. 1). A general formulation is

$$\boldsymbol{h}_{t+1} = Wf(\boldsymbol{h}_t) + W_{in}\boldsymbol{x}_t + W_{fb}\boldsymbol{y}_t, \tag{1}$$

$$\boldsymbol{y}_{t+1} = W_{out}f(\boldsymbol{h}_{t+1}), \tag{2}$$

where we put a firing rate of hidden layer at the time step t by $\boldsymbol{h}_t$, an input signal by $\boldsymbol{x}_t$, the output of this RNN by $\boldsymbol{y}_t$, and weight matrices by $W_{\cdot}$. In biological-modeling literature, a continuous-time counterpart is often used:

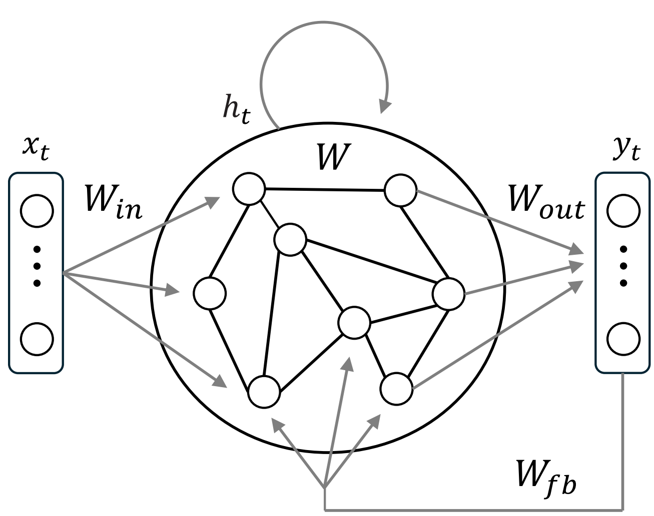

Fig. 1. Schematic diagram of recurrent neural networks.

$$\tau\frac{d\boldsymbol{h}(t)}{dt} = -\boldsymbol{h}(t) + Wf(\boldsymbol{h}(t)) + W_{in}\boldsymbol{x}(t) + W_{fb}\boldsymbol{y}(t) \tag{3}$$

which can be obtained by subtracting $\boldsymbol{h}_t$ from both sides of Eq. (1) and taking the infinitesimal-time limit. The time constant τ is introduced to control the response speed of neurons. Although individual studies differ slightly in formulation, such as the existence of the feedback loop, we do not go into such details here. The nonlinear function f is the activation function. One classical choice of f is a sigmoid function, which reflects the saturation of firing activity. In deep learning, the Rectified Linear Unit (ReLU) nonlinearity—linear for positive inputs and zero otherwise—has achieved great success for improving model's performance. It is noteworthy that ReLU is consistent with physiological experiments reporting a linear increase in firing activity after the threshold (Glorot et al., 2011).

The state update (1) is seemingly a simple mathematical equation, but its dynamics can be surprisingly complex and rich. One successful approach to figure out the dynamics is the theory of random neural networks (Amari, 1974, 1972; Poole et al., 2016; Sompolinsky et al., 1988), which assumes random weight matrices and a sufficiently large number of neurons. In random RNNs, as a specific amplitude of $\boldsymbol{W}$ (e.g., measured by its spectral radius) exceeds a critical value, the network dynamics transition to chaos (Sompolinsky et al., 1988). From an information-processing standpoint, stronger chaos corresponds to greater nonlinearity and hence a capacity to represent more complex functions (Dambre et al., 2012). However, chaotic signals behave like irregular noise, making it difficult to maintain a desired state as a memory trace. Thus, it is often best to tune the network near the *edge of chaos*.

The edge of chaos is not limited to neural networks, but refers to the vicinity of the critical line of the order-to-chaos transition in general. In this intermediate regime between rigid and irregular states, a system's computational capability is hypothesized to be maximized (Langton, 1990). In the context of RNNs, the quantification and validity of high information-processing capacity at the edge of chaos have been extensively investigated (Bertschinger, Natschläger, 2004; Boedecker et al., 2012).

In RNNs, one needs to pay attention not only to the stability of forward signal propagation but also to the stability of learning. For the training of artificial neural networks, derivatives of loss function give rise to terms of the form $\partial x_t/\partial x_1$. By the chain rule of differentiation, this expands to $\partial x_t/\partial x_{t-1} \bullet \partial x_{t-1}/\partial x_{t-2} \bullet\bullet\bullet \partial x_2/\partial x_1$, which is computed via backpropagation through time (BPTT) (Bengio et al., 1994; Pascanu et al., 2012; Werbos, 1988). Central challenges arise when training the RNN over a long-time period; the product of these gradients tends to either vanish or explode, a phenomenon known as the vanishing/exploding gradient problem. In more detail, theoretical analyses of random neural networks have revealed that the large norm of W induces exploding gradients, which typically corresponds to the chaotic phase, whereas the small norm induces vanishing gradients. Thus, to achieve stable learning dynamics, it is rational to set the network near the edge of chaos (Chen et al., 2018). Moreover, in RNNs, there is another scenario in which bifurcations inducing hysteresis in the state dynamics may appear and destabilize the learning dynamics (Doya, 2003).

It is also noteworthy that the backpropagation algorithm, which the modern machine learning of artificial neural networks strongly depends on, is nontrivial to implement in biological neural circuits and lacks clear biological plausibility. Accordingly, alternative biologically plausible training schemes have been explored. Predictive coding (Millidge et al., 2022) and equilibrium propagation (Scellier and Bengio, 2017) consider neural firing states and their update rules, which are referred to as inference, and train weight parameters so as to reduce the error information propagated through the state updates. They can be understood as approximating the gradient of the global loss function (Ishikawa et al., 2025; Millidge et al., 2023; Rosenbaum, 2022). Other methods introduce a separate backward pass specialized for the propagation of error signals; feedback alignment employs a backward pass with random weights (Lillicrap et al., 2016), while target propagation trains the backward weights via auto-encoding in a local manner (Bengio, 2014). Eligibility propagation (e-prop) approximates the BPTT gradient by the product between an eligibility trace, computed through the forward pass in a biologically plausible manner, and a global error signal, thereby avoiding explicit time-backward signal propagation (Bellec et al., 2020).

Several approaches avoid the difficulties of signal propagation and learning in RNNs. The Hopfield network (Hopfield, 1982) is a classic example: by defining an energy (Lyapunov) function that the state dynamics monotonically decreases, the state eventually settles into a memory pattern encoded in a Hebbian weight W. In other words, the state dynamics of classical Hopfield networks can avoid explosive or chaotic evolution and eventually converge to fixed points embedded as stable attractors. More recently, energy functions have been developed to cover various nonlinear activation functions and architectures including attention-based mechanisms, forming the family of modern Hopfield networks (Krotov and Hopfield, 2021; Ota and Karakida, 2023).

Another approach is the echo state network (ESN) (Jaeger and Haas, 2004), which fixes the recurrent weight W at random and trains only the readout weight W_{out}, thereby avoiding the vanishing/exploding gradient problem. In the limit of infinitely many neurons, this random RNN behaves in a form of the kernel method (Hermans and Schrauwen, 2012; Yang, 2019), so even a random weight can yield sufficient predictive performance. Although we focus here on rate-based models, it is worth mentioning that the spiking-network counterpart is known as the liquid state machine (LSN) (Maass et al., 2002). Note that ESN and LSN belong to a more general framework known as reservoir computing (RC) and represent its classical examples. In RC, a nonlinear dynamical system, not limited to neural networks, maps the input into a hidden feature space referred to as the reservoir, while typically only the readout weights are trained to predict the output signals. While the ESN is typically tuned to avoid chaotic dynamics, the first-order reduced and controlled error (FORCE) learning starts form a chaotic state and combines the state update (1) with an online training of the feedback weight W_{fb} (Sussillo and Abbott, 2009). The FORCE learning enables the network to generate input signals autonomously.

From the perspective of machine learning, approaches have also been explored that alleviate the difficulties of signal propagation and learning. The first focuses on algorithmic design such as weight clipping or teacher forcing (Pascanu et al., 2012). The second focuses on architectural improvement, most famously the long short-term memory (LSTM) (Hochreiter and Schmidhuber, 1997) and its simplification known as the gated RNN (Chung et al., 2014). By introducing multiplicative gates in the forward signal path, these architectures can selectively propagate either the raw previous state or its nonlinear transformation. Providing a direct path for the previous state mitigates vanishing or exploding gradients and enables more stable learning over long timescales (Chen et al., 2018).

3. Comparison of natural and artificial neural networks

This section summarizes how researchers typically compare biological brain networks with ANNs for elucidating the neural mechanism of decision making. In systems neuroscience, researchers record neural activity of animals performing a task and examine (i) the function of individual neurons to identify subpopulations and (ii) the low-dimensional structure of populational-level dynamics, often using dynamical systems reconstruction (Dubreuil et al., 2022; Durstewitz et al., 2023). The latter approach has recently gained attention as the number of recorded neurons increases with high-density silicon probes (Allen et al., 2019; Jun et al., 2017; Steinmetz et al., 2019) and wide-field two-photon microscopes (Ota et al., 2021; Rumyantsev et al., 2020; Sofroniew et al., 2016; Stirman et al., 2016; Stringer et al., 2021; Yu et al., 2021). Analyses of neural activity typically involve both encoding and decoding to characterize neural representations and the ability to predict task and internal variables (Funamizu et al., 2023).

Similar encoding and decoding methods are used for the analyses in artificial units in ANNs (Fig. 2). First, the ANNs are trained or designed to replicate the choice behavior observed in animals or humans during decision-making tasks by achieving the maximum outcomes (Carnevale et al., 2015; Hattori et al., 2023; Hattori and Komiyama, 2022; Mante et al., 2013). Usually, the outputs of ANNs are discretized choices during task for a simplification (Rajan et al., 2016), whereas recent studies have begun modeling rodent body movements on continuous time scales (Aldarondo et al., 2024; Ueoka et al., 2024). The activity of single artificial units (Wang, 2002) or population-level activity traces (Mante et al., 2013; Voigts et al., 2025) are then compared with biological data.

4. Recurrent neural networks for modeling short-term memory in the cortex

For modeling cortical circuits involved in decision making, we first review how RNNs model the activity for short-term memory in the cerebral cortices of monkeys and humans (Amit and Brunel, 1997; Camperi and Wang, 1998; Tsuda et al., 2023). In the delayed matching-to-sample (DMS) task (Miyashita, 1988; Miyashita and Chang, 1988), monkeys receive two successive visual stimuli separated by a several-seconds delay. The monkeys are required memorizing the first stimulus for determining whether the second stimulus is identical or different, indicating their choices by pressing a lever or touching a visual screen. These studies have identified that the neurons in temporal cortex increased the activity during the interval between the first and second stimuli for encoding the short-term memory.

In the delayed-response task, also known as the working-memory (WM) task (Funahashi et al., 1989; Funahashi and Kubota, 1994), monkeys fixate on a central spot (fixation cue) and are briefly presented a peripheral cue. After the peripheral cue disappears and a delay of several seconds, the monkeys make a saccade to the position where the peripheral cue was presented. Researchers found that neurons in the PFC near the principal sulcus increased the activity between the peripheral cue and the saccade, reflecting the memory of spatial locations of cues.

In both the DMS and WM paradigms, neurons in the cerebral cortices showed sustained activity through the delay interval. Such sustained activity for memorizing stimulus features has been modeled using RNNs with attractor dynamics (Brunel, 1996; Compte et al., 2000). These RNNs used integrate-and-fire neurons consisted of both the excitatory and inhibitory synapses (Brunel and Wang, 2001). The sustained activity of RNN relies on synaptic potentiation with long-term plasticity (LTP) (Amit and Brunel, 1997), as well as voltage-dependent excitatory postsynaptic potentials (EPSPs) produced by NMDA receptors (Lisman et al., 1998; Wang, 1999).

In contrast to the sustained activity observed in the cortex of monkeys, recent studies of mice show that the neurons in PPC have a sequence of populational activity during the delay phase of a WM decision task (Harvey et al., 2012). In this task, head-fixed mice were presented visual cues in the initial section of a virtual T-maze (Kira et al., 2023). The initial visual cues indicate whether, after a delay, the mice turn left or right at T-intersection to obtain water reward. Interestingly, such sequential activity can also be reproduced using RNN models (Rajan et al., 2016). Specifically, Rajan et al. proposed Partial In-Network Training (PINning), which uses random networks with excitatory and inhibitory connections in a rate-based neuron model. PINning changes an activity in chaotic state to a specific sequence of activity when it receives external sensory inputs. RNNs thus offer a framework for modeling short-term memory within cortical circuits for both sustained and sequential firing patterns.

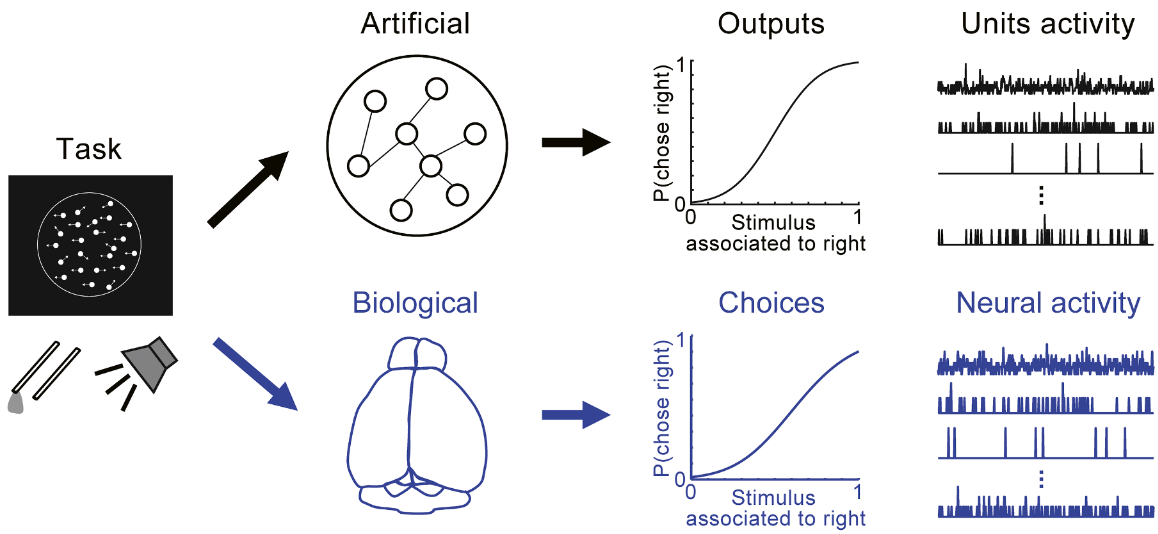

Fig. 2. Comparison of natural and artificial neural networks in decision making. Researchers first generally perform a behavioral task in animals or humans. For biological side, researchers measure the choice behavior and the brain activity in subjects during the task. Researchers also use and train artificial neural networks (ANNs) to maximize outcome in the same task. The activity of the brain and ANNs are then compared in terms of individual units or low-dimensional populational dynamics.

5. Recurrent neural networks for perceptual decision making

RNNs have been used to model cortical circuits in perceptual decision-making tasks, which require animals or humans to make choices based on uncertain sensory stimuli (Lo and Wang, 2006; Wang, 2002). Seminal examples use random dot motion (RDM) task (Gold and Shadlen, 2007), in which monkeys detect the motion direction of varying proportions of coherently moving dots. Early studies of the RDM task recorded neural activity of the middle temporal visual cortex (MT) (Britten et al., 1992; Movshon and Newsome, 1996) and the association area of lateral intro-parietal area (LIP) (Roitman and Shadlen, 2002). These studies found that the neurons in MT exhibit the motion direction of RDM, whereas the neurons in LIP integrate the direction of visual stimulus over time to form a decision. The roles of MT and LIP in the RDM task were further tested with a micro-stimulation of brain of the monkeys (Ditterich et al., 2003; Hanks et al., 2006); stimulating MT or LIP alters the speed of making choices but does not directly evoke actions, suggesting the involvements of both regions in decision-making process. The hierarchical accumulation of motion evidence, progressing from visual to association cortices, has typically been modeled using race models or drift-diffusion models (Mazurek et al., 2003). These models explain the neural activity of LIP and predict choice behaviors across various stimulus probabilities (Hanks et al., 2011) and speed-accuracy tradeoffs (Mendonça et al., 2020).

Wang and colleagues used RNNs to replicate the neural activity of LIP and the drift diffusion model with an artificial network (Lo and Wang, 2006; Wang, 2002). The RNN consisted of a pre-structured network with integrate-and-fire neurons. This network contained two populations of pyramidal neurons, each characterized by recurrent excitatory connections and competing via feedback inhibition from interneurons. Independent inputs corresponding to the MT activity were provided to each population (Britten et al., 1992). The RNN successfully captured the slow accumulation of activity observed in the LIP, suggesting that recurrent connections mediated by NMDA receptors may be involved in the integrative mechanism (Wang, 2002). Such neural activity of perceptual decisions may subsequently propagate to the basal ganglia and superior colliculus, and guide decisions with saccades (Lo and Wang, 2006). Not only these structured networks, randomly connected networks with rate-based units have also been shown to replicate the premotor cortical activity of monkeys during a tactile perceptual decision-making task (Carnevale et al., 2015). In this network, only the output weights of recurrent network were trained using the FORCE algorithm (Sussillo and Abbott, 2009). The training of feedback connections from outputs to the recurrent network changes the activity of recurrent network. Similar to the RDM task in monkeys, evidence-accumulation tasks have been investigated in head-fixed mice (Morcos and Harvey, 2016; Pinto et al., 2019). A two-module multi-task RNN with frontal and posterior modules successfully captured both the choice behaviors of mice and the neural activity patterns in dorsal cortical regions.

Mante et al. introduced contextual cues and developed a context-dependent perceptual decision-making task for monkeys (Mante et al., 2013). Subjects received a visual cue, followed by mixtures of red and green RDM stimuli. Depending on the preceding cue, subjects were asked to discriminate either the dominant color (red or green) or the dominant direction (left or right) in RDM stimuli. Even though the RDM stimulus was identical across conditions, monkeys flexibly altered their choices based on the contextual cue. Neural recordings from the PFC found heterogeneous activity of single neurons, while the population-level dynamics matched those of artificial RNN units (Mante et al., 2013). The RNN was a fully connected network with 100 units. Subsequent studies further found similarities between the activity of the PFC and the artificial units in RNN (Dubreuil et al., 2022; Langdon and Engel, 2025). Taken together, these studies show the relationship between the cortical circuits and RNN-based models in perceptual decision making.

6. Recurrent neural networks for value-based decision making

In this review, value-based decisions refer to the process where subjects make choices based on expected rewards for candidate actions (i.e., action values). These strategies are often classified as model-free or model-based (Bartolo and Averbeck, 2021; Daw et al., 2011), linked to habitual and goal-directed behaviors, respectively (Balleine, 2005; Balleine et al., 2007; Daw et al., 2005; Yin et al., 2004). In addition, inference-based strategies allow subjects to estimate the current context or the hidden state based on sensory inputs and an internal model of state transitions to make flexible decisions (Hampton et al., 2006; Vertechi et al., 2020).

Substantial studies show that basal ganglia circuits, including the striatum and dopamine neurons, represent action values and reward prediction errors (RPEs) in model-free RL (Cai et al., 2024; Cohen et al., 2012; Dabney et al., 2020; Samejima, 2005; Schultz et al., 1997; Starkweather and Uchida, 2021). The dorsolateral striatum is essential for model-free habit formation (Yin et al., 2004), whereas the dorsomedial striatum and a part of frontal cortex, i.e., the prelimbic cortex, are essential for model-based strategy (Corbit and Balleine, 2003; Yin et al., 2005a, 2005b). Moreover both human and animal studies indicate that the cerebral cortex contributes to inference-based strategies (Akam et al., 2021; Bartolo and Averbeck, 2020; Hampton et al., 2006; Vertechi et al., 2020), predicting its roles in value-based decision making (Sul et al., 2011). Taken together, these findings from the neuroscience support the hypothesize that parallel neural circuits implement multiple behavioral strategies (Daw et al., 2005; Gläscher et al., 2010). Specifically, model-free strategies update values through RPEs, whereas model-based and inference-based strategies may use state prediction errors (Bell et al., 2016) or inferences of current conditions for making choices (Yoshida and Ishii, 2006). This inference step may be independent from dopaminergic RPEs (Blanco-Pozo et al., 2024), further distinguishing the neural circuits for inference-based strategies from model-free computations.

Modeling value-based decision making with ANNs has gained attention, particularly following the success of deep RL outperforming human players in Go and other games (Schrittwieser et al., 2020; Silver et al., 2016). For example, Wang et al. proposed an RNN framework for modeling the function and circuit of PFC, termed meta-RL (Botvinick et al., 2020; Wang et al., 2018). The meta-RL hypothesizes that synaptic weights in recurrent network are modulated with dopaminergic RPEs. In the model, the artificial units are LSTM modules. The approach is considered "meta"-RL because the recurrent network itself performs value-based decision making through its internal dynamics, incorporating the exploitation-exploration trade-off, without requiring adjustments to its synaptic weights. Instead, weight updates serve to learn the structure of the task. Consequently, the meta-RL contains two hierarchical RL components: (1) the recurrent dynamics of the prefrontal network and (2) plasticity-dependent updates of the synaptic weights.

Wang et al. demonstrated that meta-RL not only replicates behavior with model-free strategy, which simply updates expected rewards of choices with RPEs (Wang et al., 2018), but also captures behavior with inference-based or model-based strategies with RPEs. These findings provide a hypothesis that RPEs become important for both the model-free and model-based behaviors, although further researches are needed for clarifying how different brain regions with different structures contribute to distinct computational roles (Doya, 1999) and whether RPEs alone can enable rapid learning of hidden task structures and state representations.

Physiological evidence for meta-RL has been observed in the orbitofrontal cortex (OFC) of mice (Hattori et al., 2023). The synaptic weight plasticity of meta-RL corresponded to the task learning across multiple sessions in mice, as demonstrated by the learning deficits with optogenetic suppression of the OFC synaptic plasticity. Moreover, lesion of the OFC affected the firing of dopamine neurons during model-based choice behavior (Takahashi et al., 2011), further supporting the links among meta-RL, the OFC, and model-based decision-making.

An updated meta-RL has been used to model hippocampal planning of action sequences and the replay of neural activity (Jensen et al., 2024). Since the meta-RL is relatively a novel technique compared to traditional RL algorithms (Sutton and Barto, 2018), further studies are required to clarify how recurrent micro circuits in cortical columns contribute to value-based decisions, and which biological mechanisms of cortical synaptic plasticity correspond to weight updates in RNNs (Goudar et al., 2023).

7. Real-cyber hybrid neural networks for developing brain-like AI

This review explained how RNNs capture the neural dynamics and functions of cerebral cortex during decision-making tasks. RNNs not only generate discrete choices in behavioral tasks but also replicate continuous body movements in animals (Aldarondo et al., 2024). For understanding the biological decision-making circuits with artificial neural networks, researchers typically first design ANNs to perform specific tasks and then compare their activity patterns with those observed in the brain (Fig. 2). Such approaches first analyze biological and artificial systems independently, and they have contributed to understanding the decision-making in the brain and to developing brain-like ANNs.

In contrast, we recently proposed a method that integrates real and artificial units from both the brain and AI, named as a real-cyber hybrid neural network (HNN) (Fig. 3a) (Ueoka et al., 2024). The concept of the HNN is to predict the activity of unrecorded brain regions, required for decision making, from recorded neuronal activity. The unrecorded activity corresponds to the artificial units in the HNN. Although recent technological advances have dramatically increased the number of simultaneously recorded neurons (Findling et al., 2023; Jun et al., 2017; Rumyantsev et al., 2020; Stringer et al., 2021), neuroscience techniques still cannot capture the neuronal activity of entire brain. Consequently, it is crucial to predict the role and activity of unrecorded brain regions for understanding how the whole brain drives decision making.

In the study of Ueoka et al., we employed mouse neuronal activity as inputs to an RNN to constrain the network dynamics with real neuronal signals (Fig. 3b). We found that the HNN outperformed conventional RNNs, which did not use mouse neuronal activity. The RNN in this study was an RC which only updated the output weights. The HNN utilized the neural activity of several cortical areas, hippocampus, or striatum. We found that the neuronal activity of most brain areas except the hippocampus improved the decoding performance of HNN in generating mouse body movements.

Further analyses of the artificial HNN units showed that temporal activity patterns of these units became similar to those of mice neurons. When we removed sound-responsive neurons of mice from the inputs of HNN, the artificial units compensated for the sound-related activity. These observations infer that the HNN approximates the neural dynamics of mouse brain compared to conventional RNNs and improves the predictions of body movements.

Another study implemented integrative feedback loops between the brain and AI to identify optimal visual stimuli that activate neurons in mouse primary visual cortex (Chen and Pesaran, 2021; Walker et al., 2019; Wang et al., 2025). ANN trained with a large-scale dataset in mouse visual cortex provides a foundation model of how a visual-cortical neuron responds to sensory stimuli (Wang et al., 2025). ANNs are used to deconvolve and characterize the heterogeneity neural responses (Tolooshams et al., 2025), and predict the dynamical transformation from neural activity to behavior (Sani et al., 2024). Taken together, these studies indicate the potential of real-cyber hybrid systems to advance understanding of brain circuits for decision making and to develop brain-like AIs.

8. Conclusion and brief summary of brain models using other artificial neural networks

This review summarized how AI-based approaches were used to elucidate the neural substrate of decision making. Specifically, we described how RNNs replicate cortical circuits during tasks involving short-term memory, perceptual decisions, and value-based decisions. From a computational perspective, several researches hypothesize that the cortex may implement unsupervised learning (Doya, 1999; van Hateren and Ruderman, 1998), sparse coding (Vinje and Gallant, 2000) or predictive coding based on Bayesian computation (Friston, 2018; Shipp et al., 2013). Indeed, some studies with animal experiments have recently proposed that the cerebral cortices implement Bayesian inference for both the value-based and perceptual decision making (Averbeck et al., 2006; Findling et al., 2023; Funamizu et al., 2016; Ishizu et al., 2024; Rao, 2010). By contrast, many of the models discussed here rely

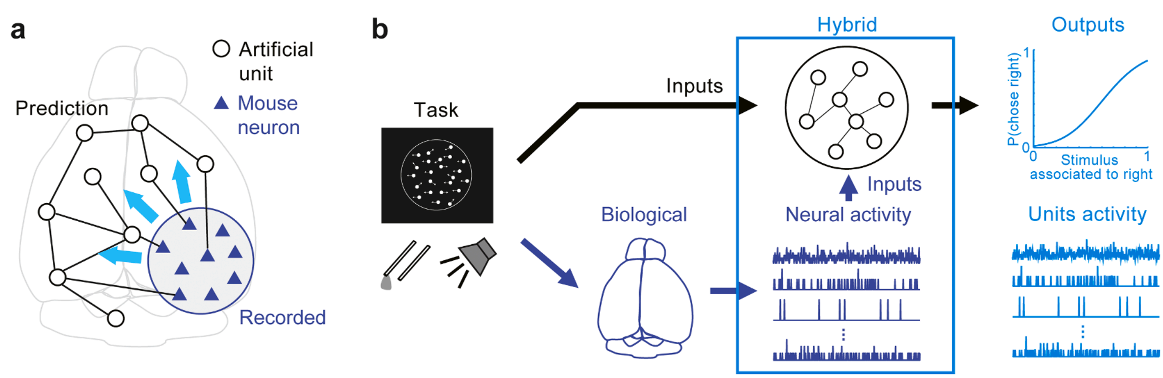

Fig. 3. Real-cyber hybrid neural network (HNN). Original data and concepts are adapted from Ueoka et al. (2024). a. Concept of the HNN. Based on recorded neural activity from animals and sensory inputs during a task, the HNN tries to predict the unmeasured neural activity of brain regions using artificial units within RNN. b. Scheme of the HNN. The HNN employs the neuronal activity of mice as inputs of RNN for generating the mice body movements. The HNN performed better than standard RNNs which did not use the neuronal activity of mice.

on supervised methods for training RNNs (Carnevale et al., 2015; Ji-An et al., 2025; Mante et al., 2013). It is interesting to elucidate how conventional machine-learning approaches relate to the recent RNN-based models for understanding the computation algorithms in the cortex.

One important piece of future research is to investigate the interplay between predictive coding and RNNs. Predictive coding frameworks typically assume that structured cortical columns and layers serve distinct functions (Shipp, 2007; Shipp et al., 2013), whereas the RNN models discussed in this review often employ random connectivity. Since some existing models already incorporate microstructures of cortical circuits, including specific inhibitory neuron subtypes (Ding et al., 2024; Wang et al., 2004), it will be informative to examine whether multi-layered neural networks with recurrent connectivity can capture both the observed neural-activity patterns and Bayesian computational principles in the cortex.

Finally, it is crucial for neuroscientists to continue assessing whether RNNs represent the most suitable models for cortical networks. Recent AI developments suggest potential links between transformer architectures and hippocampal circuits (Whittington et al., 2021), or between glia-neuron networks (Kozachkov et al., 2023). Collaborations between experimentalists and theorists are now essential for further understanding the neural mechanism of decision making.

CRediT authorship contribution statement

Ryo Karakida: Writing – original draft, Conceptualization. **Akihiro Funamizu:** Writing – original draft, Conceptualization.

Declaration of Competing Interest

The authors declare no conflict of interest.

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